Learning from Rendering: Realistic and Controllable Extreme Rainy Image Synthesis for Autonomous Driving Simulation

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Autonomous driving simulators provide an effective and low-cost alternative for evaluating or enhancing visual perception models. However, the reliability of evaluation depends on the diversity and realism of the generated scenes. Extreme weather conditions, particularly extreme rainfalls, are rare and costly to capture in real-world settings. While simulated environments can help address this limitation, existing rainy image synthesizers often suffer from poor controllability over illumination and limited realism, which significantly undermines the effectiveness of the model evaluation. To that end, we propose a learning-from-rendering rainy image synthesizer, which combines the benefits of the realism of rendering-based methods and the controllability of learning-based methods. To validate the effectiveness and generalizability of our extreme rainy image synthesizer on the semantic segmentation task, a continuous set of pixel-accurately labeled extreme rainy images is necessary. By integrating the proposed synthesizer with the CARLA driving simulator, we develop CARLARain—an extreme rainy street scene simulator which can obtain paired rainy-clean images and labels under complex illumination conditions. Qualitative and quantitative experiments validate that CARLARain can effectively improve the accuracy of semantic segmentation models in extreme rainy scenes, with the models' accuracy (mIoU) improved by 5–8% on the synthetic dataset and significantly enhanced in real extreme rainy scenarios under complex illuminations. Our source code and datasets are available at https://kb824999404.github.io/HRIG/.

CCS Concepts: • **Computing methodologies** → **Image and video acquisition**; **Scene understanding**; **Simulation support systems**;

Additional Key Words and Phrases: Rainy image synthesis, Autonomous driving, Simulation, Visual perception, Image deraining

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1 Introduction

In recent years, visual perception models [5, 17, 21, 47] have advanced significantly in autonomous driving. The performance of visual perception models for autonomous driving systems relies on the diversity of datasets. However, due to the rare occurrence of extreme weather [39, 42], the real-world efficacy of visual perception models is significantly impaired under such conditions. Extreme rainfall [48] (rain intensity $\geq$ 50 mm/h) is the most frequently occurring of extreme weather, which seriously interferes with the acquisition of scene environment information by visual perception models, thereby undermining the accuracy of tasks such as semantic segmentation, instance segmentation, and so on.

To enhance the accuracy of visual perception models, it is crucial to include a sufficient number of extreme rainy images. However, acquiring labeled real extreme rainy images is challenging due to their high costs of collection and annotation, as well as the rarity of extreme rainfalls. Therefore, a more efficient alternative is to synthesize rainy images [14, 16, 45] along with corresponding labels [9, 41].

However, existing synthetic datasets and synthesizers primarily focus on daytime scenes but lack extreme rainy scenes under complex illumination conditions, such as nighttime. As presented in Figure 1, Rain100L [49] and RainCityscapes [19] datasets only contain rainy images in daytime, while BDD350 [23] and COCO350 [23] datasets contain rainy images in both daytime and nighttime but neglect the influence of scene illumination and depth, resulting in significant color appearance differences compared to real rainy images.

Existing rainy image synthesis methods mainly fall into rendering-based methods [14, 16] and learning-based methods [32, 45, 50, 57]. Rendering-based methods, which focus on modeling rain streak appearance to render realistic rainy scenes and have advantages in realism, are limited in controllability by complex input data and empirical parameters. Learning-based methods, which train generative models on rainy image datasets to controllably generate rainy images with rain attributes (intensity, direction, etc.) and offer advantages in controllability, typically neglect essential rain attributes like color and optical effects, leading to unrealistic rain color appearances. Consequently, the inability of existing rainy image synthesis methods to balance realism and controllability leads to the inadequacy in the data diversity of extreme rainy images, especially in terms of illumination conditions.

To that end, we propose a learning-from-rendering rainy image synthesizer, which combines the benefits of realism and controllability. In the rendering stage, we propose a 3D rainy scene rendering pipeline to render a realistic, high-resolution paired rainy-clean images dataset. In the learning stage, the rendered dataset is used to train a **High-Resolution Rainy Image Generation Network (HRIGNet)**, which can capture the illumination information from the clean background images and conveniently generate extreme rainy images under the same illumination conditions. To controllably generate rain streak images for HRIGNet, we introduce the controllable rain image generation network CRIGNet [57] (which can control rain intensity, direction, and potential attributes).

To validate the effectiveness and generalizability of our extreme rainy image synthesizer on the semantic segmentation task, a continuous set of pixel-accurately labeled extreme rainy images is

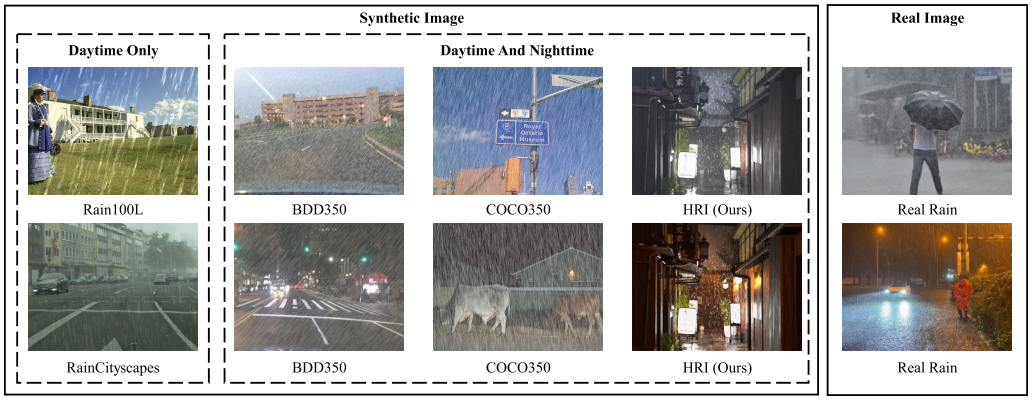


Fig. 1. Visual comparisons of real rainy images and synthetic rainy images from four existing datasets and our datasets. Rain100L [49] and RainCityscapes [19] datasets only contain rainy images in daytime. BDD350 [23] and COCO350 [23] datasets contain rainy images in both daytime and nighttime, but they neglect the influence of scene illumination and depth. Our HRI datasets contain rainy images under complex illumination conditions. Real Rain is captured in real rainy scenes. HRI, High-Resolution Rainy Image.

necessary. Given the significant challenges in collecting and labelling real-world extreme rainy images, we adopt CARLA [9], an urban driving simulator widely adopted for its open source code, protocols, digital assets, and flexible configuration of sensor suites and environmental conditions. Specifically, we leverage CARLA as a foundational framework and integrate our proposed HRIGNet into it to construct CARLARain, an extreme rainy street scene simulator. CARLARain generates extreme rainy street scenes, producing paired rainy-clean images and labels under complex illumination conditions. Unlike CARLA's built-in weather settings, which lack the capability to generate such extreme rain with fine-grained illumination control, our approach leverages HRIGNet to achieve this functionality.

We perform augmented training on existing semantic segmentation models using data generated by HRIGNet (via the CARLARain simulator). Specifically, we conduct qualitative and quantitative evaluations on synthetic and real rainy street scene datasets, focusing on validating the domain adaptation and generalization capability of HRIGNet. On the synthetic dataset, the accuracy (**mean Intersection over Union (mIoU**) of the models is improved by 5–8%. On the real dataset, the segmentation performance of the models is significantly improved under adverse conditions such as insufficient night lighting and obvious interference from rain streaks. Therefore, CARLARain can generate high-quality datasets, which can effectively improve the performance of semantic segmentation models in extreme rainy scenes.

In summary, our work makes the following contributions:

—We innovatively propose a learning-from-rendering rainy image synthesizer, which combines the benefits of realism and controllability. In the rendering stage, we propose a 3D rainy scene rendering pipeline to render realistic, high-resolution paired rainy-clean images. In the learning stage, we train a *HRIGNet* to controllably generate extreme rainy images conditioned on clean images.

—To provide a reliable evaluation basis for validating the effectiveness and generalization of our proposed synthesizer, we construct CARLARain, an extreme rainy street scene simulator. Specifically, we leverage the CARLA driving simulator as a foundational framework and

integrate our HRIGNet into it to generate paired extreme rainy-clean images and labels under complex illumination conditions.
—Qualitative and quantitative experiments validate that CARLARain can effectively improve the accuracy of semantic segmentation models in extreme rainy scenes, with the models' accuracy (mIoU) improved by 5–8% on the synthetic dataset and significantly enhanced in real extreme rainy scenarios under complex illuminations.

2 Related Work

2.1 Rainy Dataset Acquisition

Real rainy datasets are acquired by capturing rainy images in the real world. Li et al. [29] proposed a large-scale benchmark consisting of both synthetic and real-world rainy images of various rain types, but the real ones lack paired clean backgrounds. Quan et al. [35] mimicked real rainy scenes by spraying water, which can obtain paired rainy-clean images but is time-consuming.

Synthetic rainy datasets are generated with little or no human intervention, which are more efficient. Existing rainy image synthesis methods mainly fall into two categories: rendering-based methods and learning-based methods.

Rendering-based methods focus on modeling the oscillation of raindrops and the appearance of rain streaks. Garg and Nayar [14] first studied a rain streak appearance model for realistic rain rendering under different illumination. Based on their work, Halder et al. [16] proposed a practical, physically-based approach to render realistic rain in images. The RainCityscapes [19] dataset was created by adopting the images in the Cityscapes [7] dataset as clean background images, leveraging the rain streak appearance model. Despite being able to render visually realistic rain with explicit physical modeling, rendering-based methods typically rely on manual specification of illumination and scene-dependent parameters. Their simplified raindrop simulations limit flexible control over rain attributes such as intensity, spatial distribution, and appearance diversity, which makes them less suitable for scalable and controllable rainy dataset construction.

Learning-based methods use generative models to efficiently generate rainy images and exhibit superior controllability. Ni et al. [32] propose a GAN-based [15] network for continuous rain intensity control. Ye et al. [50] proposed a CycleGAN-based [58] framework that jointly learns rain generation and removal. Wang et al. [45] proposed a **Variational Autoencoder (VAE)**-based [25] model and obtained an interpretable rain generator. Based on their work, Zhou et al. [57] proposed CRIGNet, a controllable rain image generation network, which can explicitly control rain intensity and direction, and implicitly control other potential rain attributes. Despite offering convenient and flexible controllability in the latent space, most learning-based methods model rain as an additive or gray-scale layer without explicit physical constraints. Consequently, they struggle to capture essential rain attributes such as color variation, light scattering, and depth-dependent optical effects, leading to visually implausible rain appearance.

Besides methods specifically designed for rainy dataset acquisition, general image synthesis and image-to-image translation methods are also applicable to rainy image synthesis. Structure-driven models, such as neural probabilistic graphical models [54] and deep latent low-rank representation frameworks [53], are effective in learning structure-consistent mappings and preserving instance-specific details across visual domains. However, their reliance on strong structural assumptions makes them less suitable for modeling the highly stochastic and non-structural nature of rain and its complex optical effects. More recently, large-scale diffusion-based text-to-image models, e.g., Stable Diffusion [37] and Flux [26], further extend generic image synthesis to flexible image-to-image translation with prompt-based controllability. As advanced generative variants, SiT [30] builds upon DiT [33] with a flexible interpolant framework to enhance generation quality and sampling

efficiency, while VAR [43] develops a coarse-to-fine visual autoregressive paradigm to achieve superior synthesis performance and faster inference. Despite their strong generative capacity, these models mainly provide semantic-level control and lack task-specific, physically grounded modeling for rain, which often results in unrealistic rain appearance and poor scene-level consistency, especially under extreme rain conditions. Consequently, general image synthesis methods remain limited in achieving both physical realism and fine-grained controllability for realistic rainy dataset acquisition.

Overall, existing rainy image synthesis methods face a fundamental tradeoff between controllability and physical realism, which remains unresolved in current frameworks. To address this challenge, this article proposes a learning-from-rendering rainy image synthesizer that incorporates both controllability and realism into rainy image synthesis.

2.2 Visual Perception in Autonomous Driving

In autonomous driving visual perception, some researchers used multiple sensors and manual annotation to construct datasets with rainy images and labels. The Mapillary Vistas Dataset [31] is a large-scale street-level dataset with semantic and instance labels, captured at various weather (sunny, rainy, cloudy, foggy, and snowy). The nuScenes dataset [4], with a full suite of autonomous vehicle sensors, presents diverse scenes in spatial coverage, weather (sunny, rainy, cloudy), and lighting. The ACDC dataset [39], created for semantic segmentation on adverse visual conditions, includes images of four adverse conditions: fog, nighttime, rain, and snow.

Developing deep learning-based visual perception models requires massive annotated data for high performance. However, real data often struggles to generalize to domain shifts and outliers [22]. In contrast, synthetic data can capture the long-tailed distribution of task-specific environmental factors. Many works have demonstrated the effectiveness of synthetic data augmentation in visual perception. Li et al. [28], Alhaija et al. [1], and Zheng et al. [56] have respectively experimented with synthetic data augmentation for autonomous driving tasks such as semantic instance segmentation, object detection, and adverse weather perception.

To efficiently synthesize labeled datasets for multi-task visual perception models, driving simulators based on video games and graphics engines have emerged as powerful tools, leveraging their high-fidelity environment simulation. Shah et al. [41] presented AirSim, an **Unreal Engine (UE)**-based [40] simulator providing realistic simulations and a high-frequency physics engine for real-time HITL simulations. Similarly, Dosovitskiy et al. [9] proposed CARLA, an open source simulator for autonomous driving research that supports flexible specification of sensor suites and environmental conditions. Based on CARLA, Sun et al. [42] created SHIFT, a multi-task synthetic dataset with discrete and continuous shifts in cloudiness, rain/fog intensity, time of day, and vehicle/pedestrian density. However, these existing simulators mainly focus on daytime scenes and lack realistic simulations of extreme rainy scenes under complex illumination conditions.

By contrast, CARLA stands out for its open source code, protocols, and digital assets, coupled with flexible configuration of sensor suites and environmental conditions, which empower it to synthesize comprehensive label data and simulate diverse scenarios. Thus, to construct a continuous set of well-labeled extreme rainy images, we integrate the proposed synthesizer with CARLA and develop CARLARain, an extreme rainy street scene simulator.

Features of existing rain and street datasets or acquisition methods are summarized in Table 1, including rain types (real or synthetic), the proportion of extreme rainy images, label types, realism and controllability of synthetic rainy images.

Table 1. Features of Existing Rain and Street Datasets or Acquisition Methods

Name	Category	Rain type	Extreme rain	Label	Realism	Controbility
MPID-Real [29]	Real rain	Real	< 80%	—	—	—
RainDS [35]			< 70%	BG		
Photorealistic rendering [14]	Rendering-based rain	Synthetic	—		Realistic optical effects, realistic depth	Adjustable illumination and rain intensity
Physics-based rendering [16]			< 50%		Realistic optical effects	Fixed daytime, adjustable rain intensity
RainCityscapes [19]			< 50%		Realistic optical effects, realistic depth	
RICNet [32]	Learning-based rain		—		Realistic rain streak distribution	Adjustable rain intensity
JRGR [50]						No
VRGNet [45]						Adjustable latent attributes
CRIGNet [57]						Adjustable rain intensity, direction and latent attributes
Mapillary Vistas [31]	Real street	Real	< 10%	SS, IS	—	—
nuScenes [4]				2D/3D BBox		
ACDC [39]				SS		
AirSim [41]	Street simulator	Synthetic	0%	Depth, IS, OF	Realistic scene, unrealistic rain	Adjustable rain intensity and road wetness intensity
CARLA [9]				SS, IS, Depth, 2D/3D BBox, OF, Pose		Adjustable time, rain intensity, clouds amount, puddles amount, wind intensity, and camera wetness intensity
SHIFT [42]						

BBox, bounding box; BG, background; Depth, depth map; OF, optical flow; IS, instance segmentation; Pose: camera pose; SS, semantic segmentation.

3 Learning-from-Rendering Rainy Image Synthesizer

In extreme rainfalls, images captured by cameras are disturbed by rain in the scene and suffer from degradation, such as rain streaks, raindrops, fog-like rain, and so on. In this section, we mainly focus on the phenomenon of rain streaks, which occurs when falling raindrops produce motion-blurred streaks during the exposure time of the camera. Rainy images in this article refer to images that have been degraded by rain streaks.

As mentioned above, we propose a learning-from-rendering rainy image synthesizer, as shown in Figure 2. Specifically, the synthesizer combines rendering-based and learning-based methods in

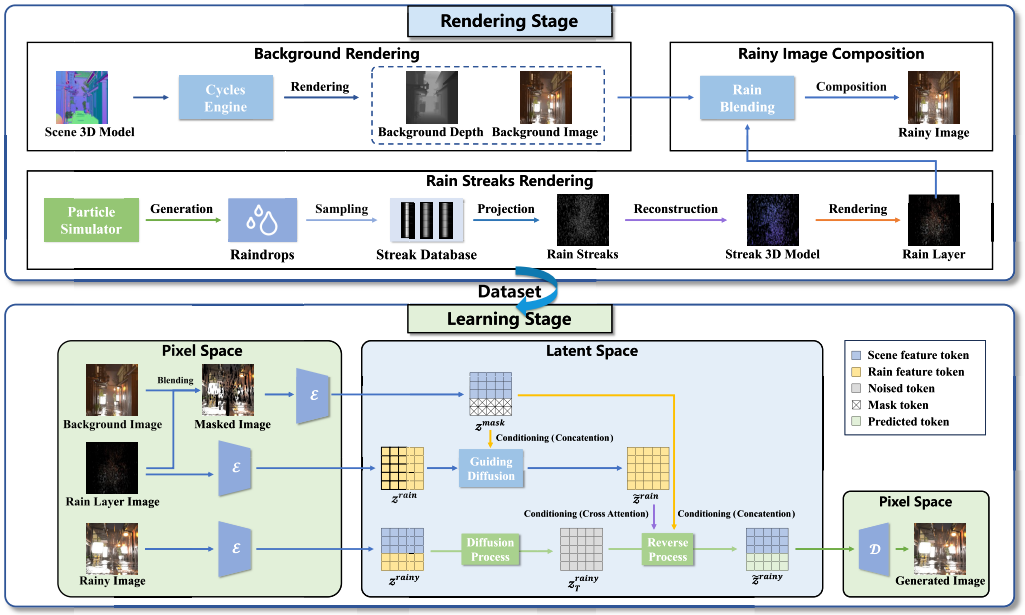


Fig. 2. Overview of our learning-from-rendering rainy image synthesizer. The top is the rendering stage, the bottom is the learning stage, and the two stages are interlinked via rainy image datasets.

two stages: the rendering stage and the learning stage. In the rendering stage, we propose a 3D rainy scene rendering pipeline to render realistic high-resolution paired rainy-clean images and create a high-resolution paired rainy-clean image dataset. In the learning stage, we train a rainy image generation network with the rendered datasets to conveniently generate rainy images under the same illumination conditions, conditioned on clean background images and rain streak images.

To further validate the effectiveness of our extreme rainy image synthesizer on the semantic segmentation task, we integrate the synthesizer with the CARLA [9] driving simulator and develop CARLARain, an extreme rainy street scene simulator which can obtain paired extreme rainy-clean images and labels under complex illumination conditions.

3.1 Rendering Stage

To train a high-quality rainy image generation network, realistic paired rainy-clean images under diverse illumination are essential. Given that offline rendering techniques based on ray tracing algorithm [34] can simulate most natural phenomena of object surface interactions and produce realistic images, they are ideal for this task. Currently, these techniques have found extensive use in the fields of movies, animation, and design. Blender [6], a popular open source 3D creation suite, has a powerful built-in unbiased path-tracer engine named Cycles, which can render realistic scene images.

In the rendering stage, we propose a 3D rainy scene rendering pipeline integrating Blender and the image-based rain rendering algorithm [14, 16]. As shown in Figure 2, it can render realistic paired rainy-background images under any illumination conditions based on the scene 3D model. Specifically, we create 3D scene models and render them with Cycles engine to obtain background RGB images and depth images. Meanwhile, we implement a raindrop physical particle simulator to generate raindrop particles, with invisible ones excluded to reduce the amount of rendering

data. Furthermore, using the rain streak rendering algorithm proposed by Garg and Nayar [14], we sample and project rain streaks from the rain streak database into the image. Then, we reconstruct each rain streak pixel into a quad to get rain streak 3D models. After merging them with the background scene 3D model, we render the resultant rainy scene 3D model for illuminated rain layers. Finally, we blend rain layers with background images to get rainy images. Each image pair thus comprises a background, a depth, a rain layer, and a rainy image.

Background Rendering. Using the Cycles engine, we render the pre-created scene 3D model for realistic background RGB and depth images. To capture full-day scenes under various illumination, we employ a procedural sky and set equal time intervals from 0 to 24 hours. For camera settings, we used focal lengths of 30–50 mm, and a 1/60 seconds shutter speed [14].

Rain Streaks Rendering. To match real-world raindrop size and distribution, we implement a raindrop particle physics simulator, referring to existing raindrop dynamics studies [3, 10, 24]. The simulation involves raindrop size, terminal velocity, and spatial distribution.[1]

To reduce rendering data, we cull invisible raindrops generated by our simulator. First, we remove raindrops outside the camera frustum. Then, using depth map and raindrop positions, we further cull occluded raindrops. Finally, we sample rain streaks from the database [14] and project onto the image following the image-based rain rendering algorithm [16].

In the image-based rain rendering algorithm [14], raindrops require scene light source illumination. To match the background scene's illumination, we generate rain streak 3D models by creating a quad at each raindrop pixel. After merging these models with the scene 3D model, we render them under the same illumination conditions as the background to obtain an illuminated rain layer.

Rainy Image Composition. We composite the background and illuminated rain layer images to generate rainy images. According to the image-based rain rendering algorithm [16], assuming that $\mathbf{x}$ is a pixel in the background image I and $\mathbf{x}'$ is the overlapping coordinate in rain layer S', the blending formula is:

$$I_{rainy}(\mathbf{x}) = \frac{T - S'_\alpha(\mathbf{x}')\tau_1}{T} I(\mathbf{x}) + S'(\mathbf{x}')\frac{\tau_1}{\tau_0}, \tag{1}$$

where $S'_\alpha(\mathbf{x}')$ is the alpha channel of the rain layer, T is the targeted exposure time, $\tau_0 = \sqrt{10^{-3}}/50$ is the time for which the raindrop remained on one pixel in the streak database, and τ_1 is the same measure according to our physical simulator.

3.2 Learning Stage

As shown in Table 2, we statistic the average image resolution of five synthetic rainy image datasets. In recent years, visual perception models have demanded increasingly higher image resolutions. For instance, Cityscapes [7]—one of the most commonly used street scene datasets—has a resolution of $2{,}048 \times 1{,}024$, yet existing synthetic rainy image datasets often fail to meet this requirement. Thus, in the learning stage, designing a network for **High-Resolution Rainy Image (HRI)** generation is crucial, as it enables the generated rainy images to match the input resolution of existing visual perception models.

To this end, in the learning stage, we propose an HRIGNet, which generates HRIs from clean background images and corresponding rain layer masks. Specifically, given an RGB background image and a mask indicating rain streak positions, HRIGNet can generate rain streaks with illumination and color consistent with the background at corresponding positions.

[1] More derivations are included in supplementary material.

Table 2. Average Resolution of Five Synthetic Rainy Image Datasets

Dataset	Image resolution
Rain100L [49]	400 × 400
COCO350 [23]	640 × 480
BDD350 [23]	1,280 × 720
RainCityscapes [19]	2,048 × 1,024
HRI (Ours)	2,048 × 1,024

Existing synthetic rainy image datasets (Rain100L, COCO350, BDD350) fail to meet the requirement of high image resolutions.

As shown in Figure 2, the architecture of HRIGNet is based on the LDM [37], where an autoencoding model [11] is employed to perceptually compress images into latent code, and the forward and reverse processes of diffusion models are conducted in the latent space.

To control the generation process of diffusion models, we adopt concatenation and cross-attention as conditional mechanisms for the reverse process. Specifically, we use a **Guiding Diffusion Model (GDM)** to predict the latent code of rain layer images, which enhances the underlying UNet backbone [38] of diffusion models via cross-attention. This provides more guidance to improve the quality of generated rainy images. To impose stronger generation constraints, we merge the background and rain layer mask into a masked image, using it as a condition for the reverse process via concatenation.

Latent Diffusion Model (LDM). Diffusion models [33, 37] are probabilistic models that gradually add noise to the data by traversing a Markov chain of T timesteps, transforming the distribution of real data to a Gaussian distribution. To reduce the training cost of high-resolution diffusion models, LDM [37] uses a pretrained VQGAN [11] autoencoding model to compress images from the high-dimensional image space to a low-dimensional latent space. This enables more efficient diffusion model training in the latent space. Given a high-resolution RGB rainy image x^{rainy}, the corresponding latent code encoded by the encoder $\mathcal{E}$ is z^{rainy}. So the objective of LDM can be expressed as:

$$L_{LDM} = \mathbb{E}_{\mathcal{E}(x^{rainy}),\epsilon\sim\mathcal{N}(0,1),t}\left[\left\|\epsilon - \epsilon_\theta(z_t^{rainy}, t)\right\|_2^2\right], \tag{2}$$

where timestep t is uniformly sampled from $\{1,\ldots,T\}$, z_t^{rainy} is a noisy version of the input z^{rainy} at timestep t, ϵ represents the noise sampled from a standard normal distribution $\mathcal{N}(0,1)$, and $\epsilon_\theta(z^{rainy}, t)$ is the denoising network's prediction of the noise component in the input data, with θ being the parameters of the model. Here, the neural backbone $\epsilon_\theta(\circ, t)$ of LDM is realized as a UNet [38].

GDM. Guiding high-resolution image generation with intermediate results provides additional information, which is expected to improve the quality of generated images. Therefore, we use a GDM trained on the latent codes of rain layer images. The GDM coarsely predicts the latent codes of the rain layer images, which are used as a condition in the reverse process of the diffusion model.

Specifically, we first merge the input RGB background and the rain layer mask into a masked image x^{mask}. Then the masked image is encoded into the latent space to obtain the latent codes

z^{mask}, which are input to the GDM to predict the latent code of rain layer image $\tilde{z}^{rain}$. The objective of GDM can be expressed as:

$$L_{GDM} = \mathbb{E}_{\mathcal{E}(x^{rain}),\epsilon\sim\mathcal{N}(0,1),t}\left[\left\|\epsilon - \epsilon_\theta(z_t^{rain}, t, z^{mask})\right\|_2^2\right], \tag{3}$$

where x^{rain} is the high-resolution RGB rain layer image, and the corresponding latent code encoded by the encoder is z^{rain}. Here, the neural backbone of GDM is realized as a UNet [38].

Conditioning Mechanisms. In the context of image generation, LDMs use the cross-attention mechanism [44] to enable inputs from different modalities to serve as conditionings. Our method leverages the predicted latent code of the rain layer image by GDM as a condition via cross-attention. Specifically, $\tilde{z}^{rain}$ is mapped to the intermediate layers of the UNet via a cross-attention layer represented as $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \cdot V$, where

$$Q = W_Q^{(i)} \cdot \varphi_i(z_t^{rainy}), K = W_K^{(i)} \cdot \tilde{z}^{rain}, V = W_V^{(i)} \cdot \tilde{z}^{rain}.$$

Here, $\varphi_i(z_t^{rainy})$ represents the intermediate representation of the denoising network ϵ_θ implemented by UNet, and $W_Q^{(i)}, W_K^{(i)}, W_V^{(i)}$ are learnable projection matrices.

Moreover, to enhance the constraints on the image generation process, we combine the concatenation and cross-attention conditioning mechanisms. We use the latent code of masked image z^{mask} as a condition of the reverse process via concatenation. Specifically, the input of the reverse process are $z_t^{concat} = [z_t^{rainy}, z^{mask}]$.

Via the concatenation and cross-attention conditioning mechanisms, we then learn the conditional LDM via:

$$L_{LDM} = \mathbb{E}_{\mathcal{E}(x^{rainy}),\epsilon\sim\mathcal{N}(0,1),t}\left[\left\|\epsilon - \epsilon_\theta(z_t^{concat}, t, \tilde{z}^{rain})\right\|_2^2\right]. \tag{4}$$

Combining the two objective functions in Equations (3) and (4), the total objective of HRIGNet is:

$$L_{HRIG} = L_{GDM} + L_{LDM}. \tag{5}$$

Image Generation. In LDM, for images of size $H \times W$ with m downsampling blocks, the input latent code to the diffusion model is of size $H/2^m \times W/2^m$, which reduces the spatial cost of training and speeds up training and inference processes. However, when m exceeds a critical value, reconstruction quality degrades, as demonstrated in [11]. Therefore, during training, we have to work patch-wise and crop images. Specifically, we train our model using images of up to 512×512 resolution. Considering that spatial conditioning information [11] is available in our model, during inference, we can simply segment background images into patches of size 512×512 and use them as conditioning. By merging the results, we can generate images up to $2{,}048 \times 1{,}024$ resolution.

In our implementation, simply segmenting background images into blocks and feeding them to the model leads to hue variations in output results, causing a blocky effect in the merged output. We address this by using rain layer masks to blend generated images and background images.

3.3 Extreme Rainy Street Scene Simulator

This article integrates our proposed HRIGNet with the CARLA [9] driving simulator and the controllable rain image generation network CRIGNet [57], constructing CARLARain, an extreme rainy street scene simulator, as shown in Figure 3. Next, we will introduce the implementation of CARLARain in detail.

Street Scene Simulation. To enable realistic physical and visual street scene simulations and obtain labeled data for visual perception models, we introduce the CARLA driving simulator. CARLA [9] is an open source urban driving simulator implemented as an open source layer over

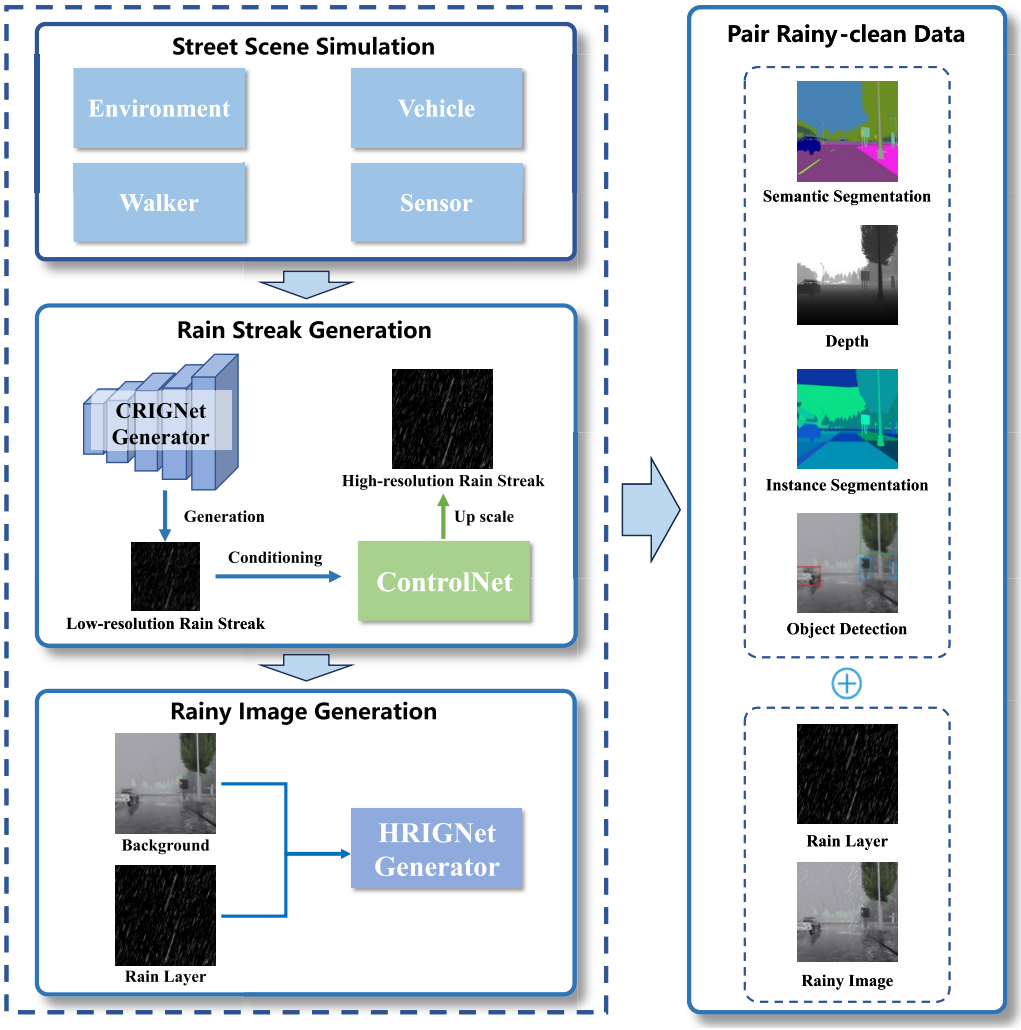


Fig. 3. Overview of our extreme rainy street scene simulator, CARLARain. By integrating the proposed HRIGNet with the CRIGNet and the CARLA, CARLARain can obtain paired extreme rainy-clean images and label information data under complex illumination conditions.

UE [27], which provides state-of-the-art rendering quality [27] and realistic physics. CARLA allows for flexible configuration of commonly used visual sensors, enabling the collection of scene images and label data during simulation. In this article, our CARLARain configures RGB camera, depth camera, semantic segmentation camera, instance segmentation camera, and collision detection sensor based on CARLA. During simulation, CARLARain can capture frame-by-frame street scene data, RGB, semantic/instance segmentation, and depth images, as well as object bounding boxes.

Rain Streak Generation. Due to the limitations of the VAE on which CRIGNet is built, it can only generate rain streak images with a maximum resolution of 256×256. To generate high-resolution

Table 3. Overview of the HRI Dataset

Scene	Dataset type	Camera	Frames	Intensities	Image pairs	Resolution
Lane	Train set	3	100	4	1,200	2,048 × 1,024
	Test set	1			400	
City street	Train set	5	25	4	500	
	Test set	1			100	
Japanese street	Train set	8	25	4	800	
	Test set	2			200	

rain streak images, we pretrain a ControlNet-based [52] up-scale model.[2] Using low-resolution rain streak images from CRIGNet as control conditions, it generates corresponding high-resolution ones. By inputting rain intensity and direction variables into CRIGNet and ControlNet, CARLARain can controllably generate rain streak images for rainy image generation.

Rainy Image Generation. After acquiring clean scene RGB images and rain streak images, CARLARain can generate realistic rainy images under the same illumination conditions as the background environment by inputting these images as conditional information into the HRIGNet. This allows CARLARain to produce paired extreme rainy-clean images and label data under complex illumination conditions.

4 Experimental Results

In this section, we conduct qualitative and quantitative experiments to validate the realism and controllability of our proposed rainy image synthesizer, as well as its effectiveness in improving the performance of semantic segmentation models in extreme rainy scenes.[3]

For rainy image generation, we compare HRIGNet with some baseline image generative models to validate its performance in HRI generation. Moreover, we conduct image deraining experiments to validate that it can improve the accuracy of image deraining models, thus indirectly validating the realism of the generated rainy images. Additionally, we demonstrate some rainy images, illustrating their controllability in attributes such as illumination, rain intensity, and direction.

Based on CARLARain, we construct an extreme rainy street scene image dataset, ExtremeRain. We test existing semantic segmentation models on it to evaluate the impact of rainy scenes under different illumination conditions. Furthermore, to improve model accuracy in extreme rainy scenes, we conduct augmented training with ExtremeRain and perform qualitative and quantitative evaluations on synthetic and real datasets.

4.1 Dataset

HRI Dataset. To train the HRIGNet, we create a HRI dataset in the rendering stage of the proposed rainy image synthesizer. The HRI dataset comprises a total of 3,200 image pairs. Each image pair comprises a clean background image, a depth image, a rain layer mask image, and a rainy image. As shown in Table 3, it contains three scenes: lane, city street, and Japanese street, with image resolutions of 2,048 × 1,024. The dataset covers three representative road environments (Japanese alley, city street, and Japanese street), multiple camera viewpoints, and four rain intensities, providing diverse spatial layouts and rain conditions for training and evaluation.[4]

ExtremeRain Dataset. As shown in Table 4, the ExtremeRain dataset contains eight different scenes and three illumination conditions: daytime, sunset, and night. The rainy scenes feature a

[2]Details are included in supplementary material.
[3]Implementation details are included in supplementary material.
[4]Details about HRI dataset are included in supplementary material.

Table 4. Overview of the ExtremeRain Dataset

Dataset type	Scene	Time	Frames	Samples	Resolution
Train set	7	3	1,000	21,000	2,048 × 1,024
Test set	1	3	1,000	3,000	

Table 5. Time Consumption of Core Components in Our Framework

Component	Task	Time consumption
Rendering-stage	Render 2,048 × 1,024 image pair	1.5 min/image pair
Learning-stage	Train HRIGNet (100 epochs)	8 h
	Inference single 2,048 × 1,024 image	1 s/image
CARLARain	Generate labeled images	0.15 s/image pair
CRIGNet	Inference 2,048 × 1,024 rain-layer image	0.02 s/image

rain intensity ranging from 5–100 mm/h, covering extreme rainfalls under complex illumination conditions. It contains labels for semantic segmentation, instance segmentation, depth estimation, and object detection. The dataset thus features diverse road environments, including urban downtown areas, highways, residential towns, mountainous regions, and rural environments, as well as three illumination conditions and a broad range of rainfall intensities, enabling reliable evaluation under complex extreme rain scenarios.[5]

Time Consumption. We conduct a time consumption analysis for all core components of our framework, including the rendering stage and the learning stage of the proposed rainy image synthesizer, as well as CARLARain for background image rendering and CRIGNet for rain layer generation. All experiments are performed on a server equipped with an Intel i9-13900K CPU, 64 GB DDR5 RAM, and an NVIDIA RTX 4090 GPU with 24 GB memory. The reported results are averaged over three independent runs and summarized in Table 5. The rendering stage takes 1.5 minutes to generate a single 2,048 × 1,024 rainy–clean image pair. For the learning stage, HRIGNet requires 8 hours for end-to-end training over 100 epochs, and only 1 seconds for single-image inference. CARLARain can render single-paired and labeled 2,048 × 1,024 images in 0.15 seconds. CRIGNet achieves an inference speed of 0.02 seconds per 2,048 × 1,024 rain-layer image.

4.2 Realism and Controllability of Rainy Image Generation

Compare with Baseline. To evaluate the performance of HRIGNet in HRI generation, we compare it with several baseline image generative models: LDM [37], DiT [33], CycleGAN [58], VAR [43], and SiT [30]. The evaluation metrics used are **Frechet Inception Distance (FID)** [18], **Learned Perceptual Image Patch Similarity (LPIPS)** [55], **Structural Similarity Index Measure (SSIM)**, and **Peak Signal-to-Noise Ratio (PSNR)**. As shown in Table 6, at the resolution of 512 × 512, our model achieves the best performance in terms of FID, LPIPS, and SSIM, while at the higher resolution of 1,024 × 1,024, HRIGNet attains the best results in FID, SSIM, and PSNR. Figure 4 illustrates a comparison of rainy image generation results of these methods. As seen, our method can well capture the illumination and color in the background image, and map them to the generated rain layer. As a result, the rain layer is endowed with a visually plausible color appearance that matches the background image.

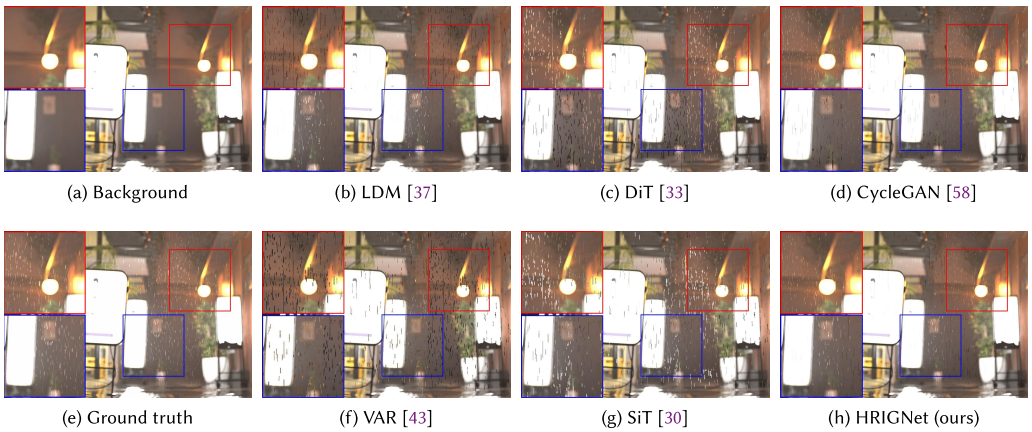
Image Deraining. With HRIGNet, sufficient rainy images can be conveniently generated from background images and rain layer masks, enabling us to create new paired rainy image datasets or

[5]Details about ExtremeRain dataset are included in supplementary material.

Table 6. Quantitative Evaluation of Baseline Models and HRIGNet

Method	Patch size	FID↓	LPIPS↓	SSIM↑	PSNR↑
LDM [37]	512 × 512	46.583	0.241	0.703	16.652
	1,024 × 1,024	25.790	0.194	0.880	28.435
DiT [33]	512 × 512	164.977	0.490	0.548	12.097
	1,024 × 1,024	67.179	0.300	0.829	23.980
CycleGAN [58]	512 × 512	47.073	0.271	0.639	**21.604**
	1,024 × 1,024	28.162	**0.157**	0.870	27.301
VAR [43]	512 × 512	71.873	0.304	0.699	20.137
	1,024 × 1,024	81.878	0.339	0.690	19.478
SiT [30]	512 × 512	54.107	0.274	0.743	21.585
	1,024 × 1,024	56.352	0.298	0.720	21.433
HRIGNet	512 × 512	**32.111**	**0.205**	**0.747**	18.595
	1,024 × 1,024	**16.060**	0.160	**0.899**	**29.406**

Bold values denote the best performance under the corresponding patch size for each evaluation metric.



(a) Background (b) LDM [37] (c) DiT [33] (d) CycleGAN [58]

(e) Ground truth (f) VAR [43] (g) SiT [30] (h) HRIGNet (ours)

Fig. 4. Visual comparison of generated rainy images from different image generative models. Our HRIGNet can better capture the illumination and color in the background image and map them to the generated rain layer.

augment existing datasets. In this subsection, we use HRIGNet to augment the existing datasets with ratio 1, further improving the deraining performance of current DL-based derainers on real rain datasets.

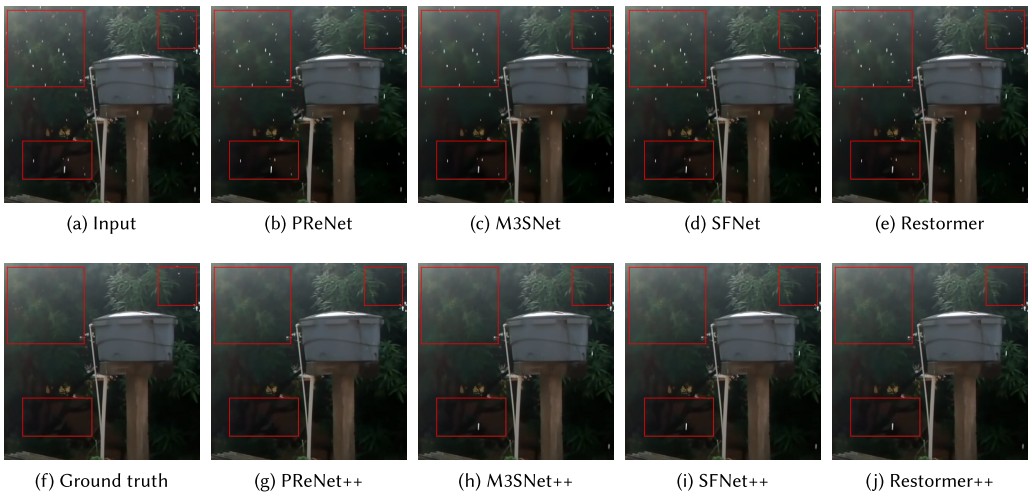
We evaluate the effectiveness of the augmentation strategy benefitted from HRIGNet with the latest DL-based SIRR methods, including PReNet [36], M3SNet [13], SFNet [8], and Restormer [51]. The train sets are common synthetic datasets, including RainTrainL [49] and Rain1400 [12]. We augment these datasets, retrain the derainers, and compare their generalization performance on the real dataset SPA-Data [46].

The quantitative comparison is shown in Table 7, where "+" denotes the augmented training with physics-based rendering [16] (where we use to produce rain layer masks for our HRIGNet) and "++" denotes the augmented training with our method. $\Delta \uparrow$ represents the performance gain brought by the augmented training with our method. As seen, our method improves the performance of all

Table 7. Quantitative Comparisons on Test Data of SPA-Data

Dataset	RainTrainL			Rain1400		
Metric	FID↓	SSIM↑	PSNR↑	FID↓	SSIM↑	PSNR↑
PReNet	45.981	0.941	33.493	48.834	0.937	31.869
PReNet+	45.586	0.942	33.168	49.001	0.934	32.157
PReNet++	**43.706**	**0.946**	**33.614**	**46.073**	**0.944**	**33.064**
Δ ↑	2.275	0.005	0.121	2.761	0.007	1.195
M3SNet	47.989	0.939	33.147	53.98	0.926	31.04
M3SNet+	49.173	0.938	33.062	49.9	0.934	32.897
M3SNet++	**46.889**	**0.943**	**33.173**	**48.609**	**0.94**	**33.097**
Δ ↑	1.100	0.004	0.026	5.371	0.014	2.057
SFNet	48.807	0.939	33.164	51.351	0.932	31.802
SFNet+	46.155	0.941	33.303	48.798	0.936	33.213
SFNet++	**44.982**	**0.946**	**33.316**	**46.852**	**0.944**	**33.205**
Δ ↑	3.825	0.007	0.152	4.499	0.012	1.403
Restormer	50.785	0.932	32.821	53.598	0.929	31.446
Restormer+	47.795	0.938	33.251	48.575	0.936	33.188
Restormer++	**46.476**	**0.943**	**33.397**	**46.57**	**0.943**	**33.292**
Δ ↑	4.309	0.011	0.576	7.028	0.014	1.846

"+" denotes the augmented training by physics-based rendering [16], "++" denotes the augmented training our method. Δ ↑ represents the performance gain brought by the augmented rains generated by our method. Bold values indicate the best performance of each metric among all compared methods on the identical dataset.



(a) Input (b) PReNet (c) M3SNet (d) SFNet (e) Restormer

(f) Ground truth (g) PReNet++ (h) M3SNet++ (i) SFNet++ (j) Restormer++

Fig. 5. Visual comparison of rain removal results on a test image from SPA-Data. The first row is the input rainy image and the output of derainers trained on the original Rain1400 train set. The second row is the ground truth background image and the output of derainers trained on the Rain1400 train set augmented by our HRIGNet.

derainers to varying degrees. Figure 5 shows the visual comparison of rain removal results on a test image from SPA-Data. Under such a complex rainy scene, these derainers with augmented training evidently remove rain. The results validate that the rainy images generated by HRIGNet can help improve the accuracy of these deep derainers to rainy images in the real world, thus indirectly validating the realism of the generated rainy images.

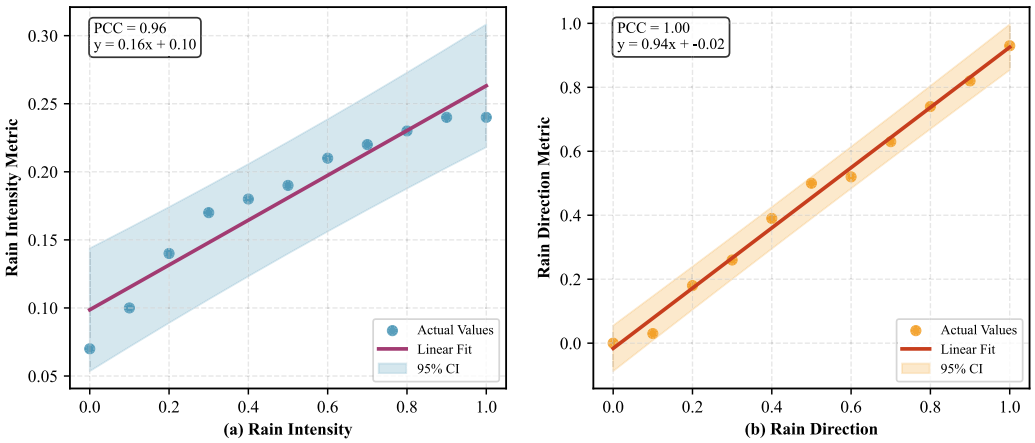


Fig. 6. Quantitative validation of rain intensity and direction controllability. The x-axis denotes the manually set rain attribute values within $[0, 1]$, and the y-axis represents the derived metrics (rain intensity: percentage of rain streak pixels or average mask intensity; rain direction: dominant gradient orientation). Scatters show the derived metric values of generated images, solid lines indicate linear fits, and shaded areas denote the 95% confidence intervals. The high PCC values (0.96 for intensity, 1.00 for direction) demonstrate an extremely strong linear correlation, validating the precise controllability of rain intensity and direction in our method.

Controllability. As shown in Figure 7, representative rainy images from ExtremeRain are presented. By combining HRIGNet with CRIGNet and CARLA, we can control different background scenes, achieve variations in illumination such as daytime, sunset, and night, and manipulate attributes like rain intensity and direction. The controllability of multiple attributes ensures the diversity of the dataset. In addition, to quantitatively validate the precise controllability of rain intensity and direction, we fix the background scene and generate smooth interpolation sequences of rainy images with the two attributes varying within $[0, 1]$. Specifically, we derive the rain intensity metric (normalized to $[0, 1]$) from the density ratio of rain streaks based on the rain streak mask output by CRIGNet, which is computed as the percentage of rain streak pixels within the mask. For rain direction, we estimate the dominant direction via gradient orientation histograms calculated within the rain streak regions, and normalize it to $[0, 1]$. We then compute the **Pearson Correlation Coefficient (PCC)** [2] between the CRIGNet input values and these derived intensity and direction metrics to quantify the degree of linear correlation. As shown in Figure 6, the PCC values of rain intensity and direction reach 0.96 and 1.00, respectively, both showing a strong linear correlation, which validates the accurate control of rain intensity and direction in our proposed method.

4.3 Semantic Segmentation in Extreme Rainfall

Semantic Segmentation Evaluation. To evaluate the impact of extreme rainy scenes under different illumination conditions, we test several SOTA semantic segmentation models (including ViT-Adapter [5], InternImage [47], SeMask [21], and DiNAT [17]) on clean and rainy RGB images from the ExtremeRain dataset, respectively. These models are all pretrained on the Cityscapes [7] dataset. Visual comparisons are shown in Figure 8, while quantitative results are presented as line charts in Figure 9. It can be seen from these results that the accuracy of all models decreases in rainy scenes. In daytime scenarios, with simple illumination conditions, rainy scenes have a minor impact on model accuracy. By contrast, in sunset and night scenarios, the dual impact of complex illumination and extreme rainfalls leads to a significant drop in semantic segmentation accuracy.

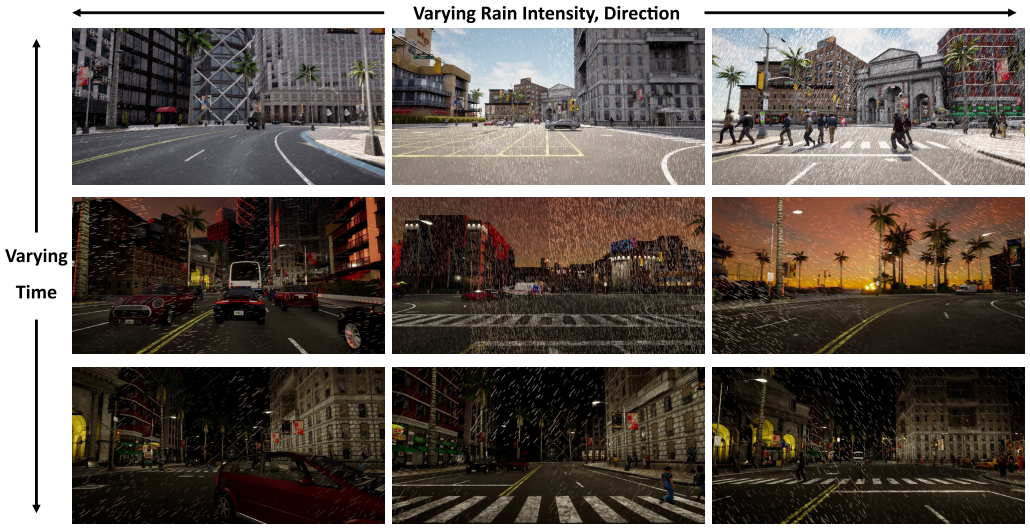


Fig. 7. Samples of rainy images from ExtremeRain with multi-attribute controllability (varying in scenes, time, rain intensity, and direction).

Semantic Segmentation Augmentation. To improve the accuracy of semantic segmentation models in extreme rainy scenes, we further conduct augmented training with the ExtremeRain dataset and evaluate the models on synthetic and real datasets.

Firstly, we select 5,000 rainy images from discrete domains of the synthetic dataset SHIFT [42] as the test set for quantitative evaluation (mIoU). We separately conduct augmented training on semantic segmentation models using background images, rainy images from the ExtremeRain dataset, and rainy images generated by mainstream image generation models (SDv1.5 [37], SDXL [37], Flux [26]), and compare their prediction results, as shown in Table 8, where "Bg-Aug" denotes augmented training with background images, "Rainy-Aug (Ours)" denotes augmented training with rainy images from ExtremeRain dataset, and "SDv1.5-Aug/SDXL-Aug/Flux-Aug" denote augmented training with rainy images generated by SDv1.5/SDXL/Flux, respectively. $\Delta \uparrow$ represents the performance gain brought by augmented training.

The experimental results show that most models trained with data augmentation achieve higher accuracy than the pretrained ones, and augmentation based on rainy images from the ExtremeRain dataset consistently outperforms background-image-based augmentation. This indicates that the original training data of the pretrained models lack sufficient diversity, particularly in extreme rainy scenarios. Among all image augmentation strategies, rainy images from our ExtremeRain dataset yield the most significant performance improvement (mIoU increases by 5–8%). However, rainy images generated by SDv1.5, SDXL, and Flux generally perform worse than background-image-based augmentation. This suggests that, despite their strong generative capacity, general image synthesis models lack physically grounded rain modeling and fine-grained controllability, leading to less realistic rain patterns and insufficient scene-level consistency, especially under extreme rain conditions. In contrast, ExtremeRain explicitly models rain formation and its optical effects, enabling the generation of more realistic and task-relevant rainy images, which in turn leads to substantially larger performance improvements in semantic segmentation.

Finally, we conduct a qualitative evaluation using real extreme rainy street scene images to validate that ExtremeRain dataset can improve the performance of semantic segmentation models in real-world extreme rainy conditions. We collect real rainy scene images with rain intensity ≥ 50

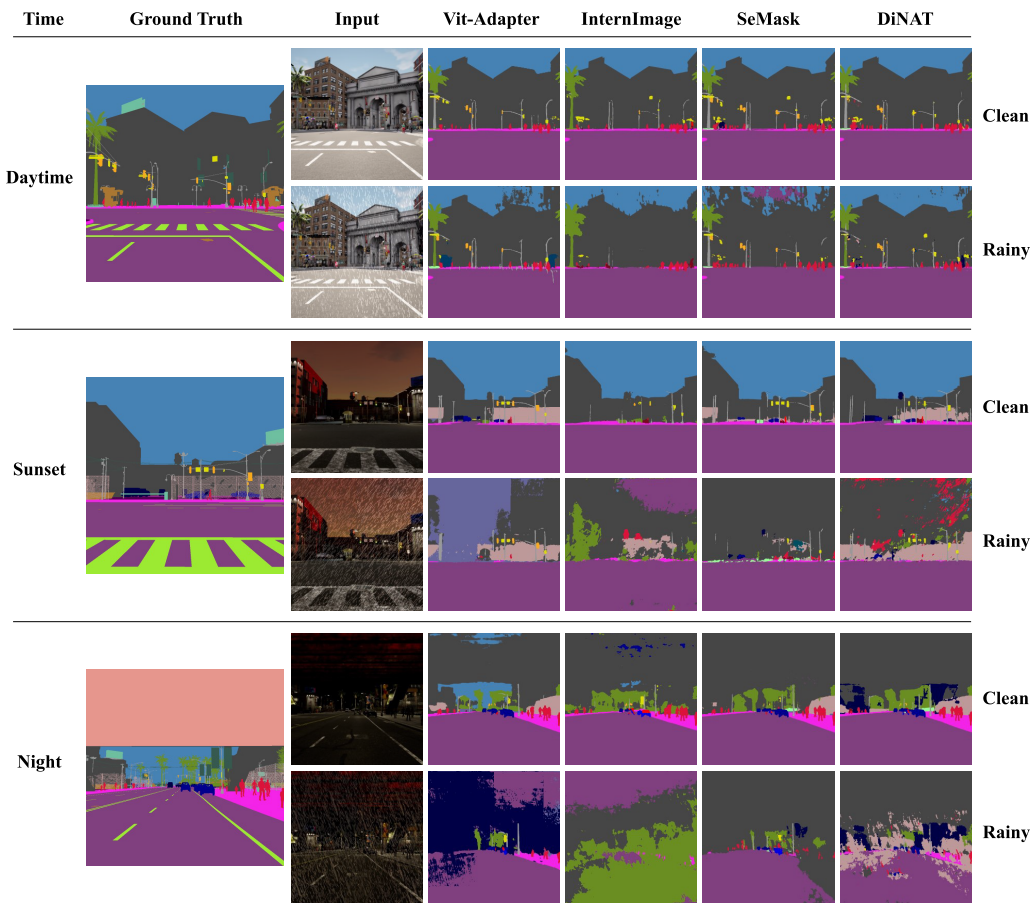


Fig. 8. Visual comparisons of semantic segmentation results on clean and rainy RGB images from ExtremeRain dataset. The accuracy of all models decreases in rainy scenes.

mm/h from the Internet, and use them as the test set. The test set covers diverse and challenging characteristics: it includes moderate to torrential rain scenarios with dense vertical or wind-induced diagonal rain streaks, and encompasses complex illumination conditions (low-light evening/night, overcast daytime) with localized highlights/shadows from artificial lighting. The scenes consist of typical urban elements (buildings, lane markings, utility poles, vehicles, pedestrians) with frequent object occlusion and reflective wet surfaces. These factors pose key challenges for semantic segmentation, including rain-induced blurring, low contrast, and contour occlusion.

As shown in Figure 10, under the above-described extreme rainy conditions, we visually compare the prediction results of pre-trained models and those augmented with rainy images on the test set. Results indicate that augmented training enables models to better recognize object contours (e.g., buildings and sky) in long-shot views and refine segmentation of nearby objects (e.g., trees and lane lines). In extreme rainy scenarios under complex illuminations, where lighting is poor and rain streaks prominent, the segmentation performance improves significantly.

To further complement the above qualitative analysis and validate the effectiveness of the ExtremeRain dataset on real-world extreme rainy scenes (where ground truth annotations are unavailable), we adopt a large Vision–Language Model (GPT-4o [20]) to conduct a standardized

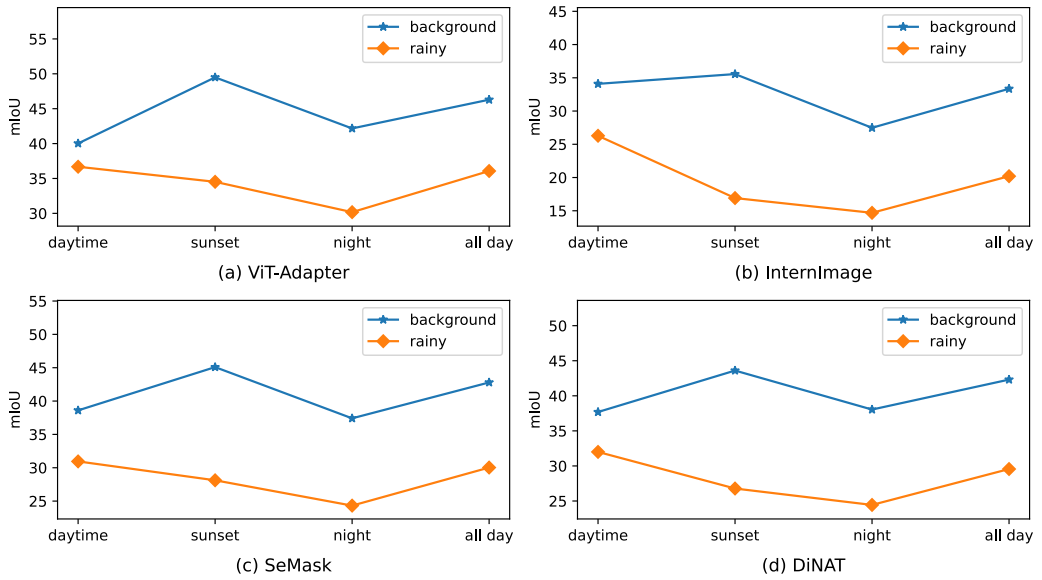


Fig. 9. Semantic segmentation quantitative results (mIoU) on clean and rainy RGB images from ExtremeRain dataset. In sunset and night scenarios, the dual impact of complex illumination and extreme rainfalls leads to a significant drop in semantic segmentation accuracy.

Table 8. Quantitative Results (mIoU) on the SHIFT Dataset

Model	Pretrained	Bg-Aug		Rainy-Aug (Ours)		SDv1.5-Aug		SDXL-Aug		Flux-Aug	
	Predict	Predict	Δ ↑	Predict	Δ ↑	Predict	Δ ↑	Predict	Δ ↑	Predict	Δ ↑
Vit-Adapter	37.38	39.55	2.17	**42.89**	**5.51**	37.47	0.09	35.74	−1.64	37.80	0.42
InternImage	30.88	34.79	3.91	**38.12**	**7.24**	31.82	0.94	27.97	−2.91	30.26	−0.62
SeMask	32.28	38.80	6.52	**40.26**	**7.98**	33.16	0.88	31.77	−0.51	35.52	3.24
DiNAT	34.86	38.14	3.28	**40.58**	**5.72**	33.10	−1.76	32.10	−2.76	37.60	2.74

Bg-Aug, Augmented training with background images; Rainy-Aug, Augmented training with rainy images from ExtremeRain Dataset; SDv1.5-Aug/SDXL-Aug/Flux-Aug, Augmented training with rainy images generated by SDv1.5/SDXL/Flux, respectively. The accuracy of all augmented models is improved by 5−8%. Bold values represent the optimal prediction mIoU for each model across all augmentation strategies.

coarse-grained quantitative scoring (1-5 points, 5 for optimal performance) on the semantic segmentation results of real rainy street scene images. The scoring covers four core dimensions: contour integrity, category accuracy, rain interference resistance, and overall consistency, with the average score of all test samples used as the evaluation metric. As shown in Table 9, all models exhibit clear performance gains: the pre-trained models yield average scores ranging from 3.57 to 3.88, while Bg-Aug (background-augmented) models reach 3.58 to 4.01, and Rainy-Aug (rain-augmented) models attain the highest scores of 4.04 to 4.26. This clearly demonstrates that Rainy-Aug outperforms Bg-Aug, and both surpass the pre-trained models, which confirms that the ExtremeRain dataset can better enhance the model's adaptability to extreme rainy scene features (e.g., rain streaks, low contrast) compared with background-only augmentation. The significant score improvement of Rainy-Aug (average increase of 0.32–0.48 points compared to pre-trained models) further verifies that our dataset effectively compensates for the lack of extreme rainy scene samples in the original training set, enabling models to maintain robust segmentation performance in real challenging rainy scenarios.

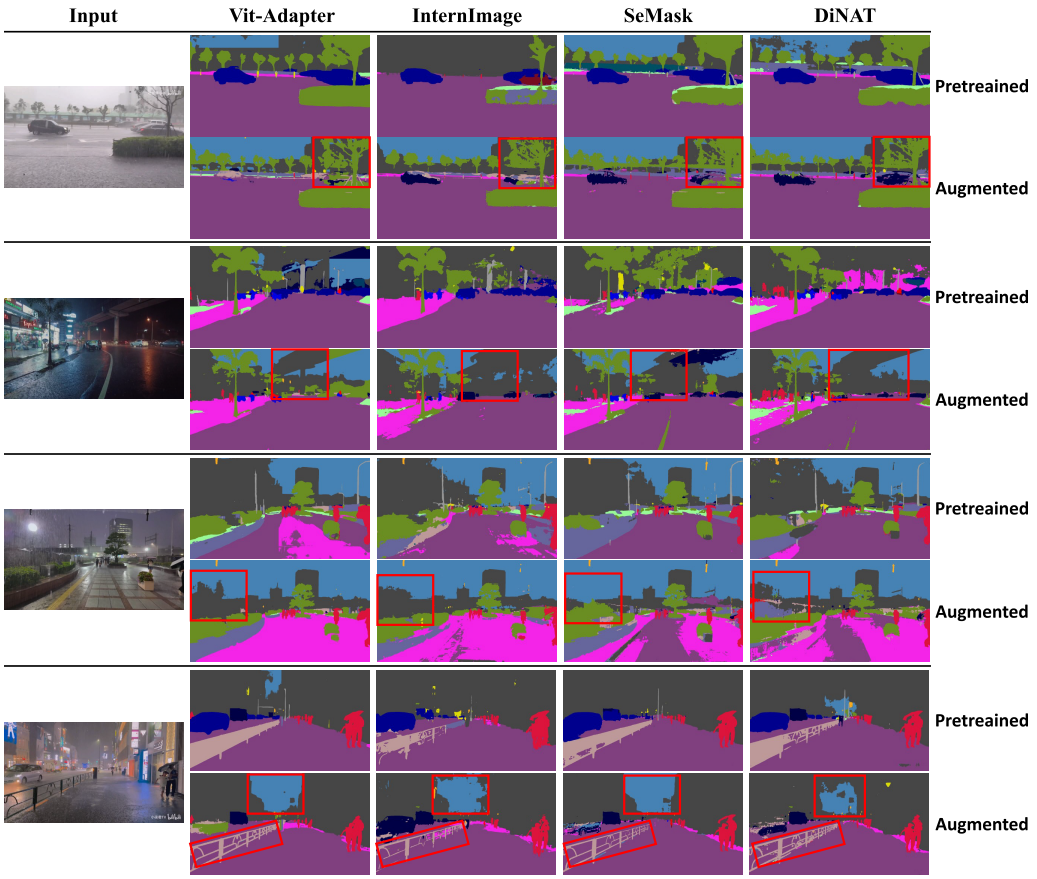


Fig. 10. Visual comparisons of semantic segmentation results on real data. "Pretrained" denotes prediction results of pre-trained models, and "Augmented" denotes those augmented with rainy images from ExtremeRain.

Table 9. Quantitative Scoring Results of Semantic Segmentation on Real Extreme Rainy Scenes (1–5 Points)

Model	Pretrained	Bg-Aug	Rainy-Aug (Ours)
Vit-Adapter	3.88	4.01	**4.26**
Intern	3.57	3.58	**4.04**
Semask	3.66	3.95	**4.14**
DiNAT	3.80	3.96	**4.12**

Bold values mean the highest segmentation score of each model across all augmentation schemes.

However, augmented training may introduce a potential bias toward rainy scene features (e.g., rain streaks), reducing prediction accuracy for objects resembling rain streaks, like utility poles, as the overemphasis on learning rain-specific patterns can blur the distinction between such features and the core visual cues of similar objects, leading to misclassification. Thus, future work should optimize the training strategy. For instance, combining our extreme rainy scene dataset with existing ones to form a new dataset and training models anew can balance their adaptability to both rainy and clean scene features.

5 Conclusion and Limitations

Existing rainy image synthesizers often suffer from poor illumination controllability and limited realism, which significantly undermines the effectiveness of the model evaluation. To that end, we propose a learning-from-rendering rainy image synthesizer, which combines the benefits of realism and controllability. By integrating the proposed synthesizer with the CARLA driving simulator, we develop CARLARain—an extreme rainy street scene simulator which can obtain paired rainy-clean images and labels under complex illumination conditions. Qualitative and quantitative experiments validate that CARLARain can effectively improve the accuracy of semantic segmentation models in extreme rainy scenes.

However, there are still avenues for further improvement. Our rainy image synthesizer currently connects the two stages through pre-collected rainy image datasets, which limits the efficiency of early dataset construction, and future work will explore a more effective integration of these stages. HRIGNet may generate overly bright rain layers under strong illumination conditions (e.g., daytime scenes); thus, we plan to leverage general image synthesis models to first translate scenes into overcast illumination before rain synthesis. Moreover, patch-wise generation for high-resolution images can introduce blocky artifacts when background illumination varies significantly, which could be alleviated through inter-patch illumination fusion. Finally, the current framework mainly focuses on rain streaks, while other dynamic rainy effects, such as splashes, vehicle wipers, and reflective surfaces, as well as a refined feedback mechanism between CARLARain and autonomous driving systems for joint optimization, remain important directions for future work.

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